Disclaimer/credit. The toy examples in this sheet have been made up, and do not represent any physical system of note. Any resemblance to real physics that you can obtain from them is entirely coincidental.

1) Classify the wave equation as:

   (a) A second-order scalar PDE,

$$\frac{\partial^2 \bar{u}}{\partial t^2} = a^2 \left(\frac{\partial^2 \bar{u}}{\partial x^2} + \frac{\partial^2 \bar{u}}{\partial y^2} \right)$$

   (b) A system of first-order PDEs,

$$\frac{\partial}{\partial t} \begin{pmatrix} \bar{u} \\ \bar{v} \end{pmatrix} + \frac{\partial}{\partial x} \begin{pmatrix} \bar{v} \\ a^2 \bar{u} \end{pmatrix} = 0$$

2) Classify the heat equation,

$$\frac{\partial \bar{u}}{\partial t} = \alpha \left(\frac{\partial^2 \bar{u}}{\partial x^2} + \frac{\partial^2 \bar{u}}{\partial y^2} \right)$$

3) For the following system of PDEs,

$$\frac{\partial}{\partial t} \begin{pmatrix} a \\ b \\ c \end{pmatrix} + \frac{\partial}{\partial x} \begin{pmatrix} a + 1.4b \\ b + \dfrac{c}{1.4} \\ 1.4b + c \end{pmatrix} = 0$$

   (a) Express this equation in non-conservative form.

   (b) Show that this is a hyperbolic system.

   (c) Find the characteristic form for this system.

4) Consider the (entirely unphysical, under dimensional analysis) system

$$\frac{\partial}{\partial t} \begin{pmatrix} r \\ s \end{pmatrix} + \frac{\partial}{\partial x} \begin{pmatrix} r^2 - s^2 \\ 3r + s \end{pmatrix} = 0$$

   For what values of r and s is this system:

   (a) elliptic?

   (b) parabolic?

5) Consider the compressible Euler equations,

$$\begin{aligned} \frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) &= 0, \\ \frac{\partial (\rho \mathbf{v})}{\partial t} + \nabla \cdot \left(\rho\, \mathbf{v} \otimes \mathbf{v} + \mathbf{I} p \right) &= 0, \\ \frac{\partial E}{\partial t} + \nabla \cdot \left[(E + p) \mathbf{v} \right] &= 0, \end{aligned}$$

   where we shall invent an equation of state (this step is completely unphysical)

$$p = \alpha E$$

   where α is a constant.

(a) Express these equations in 1D.

(b) Write the 1D equations in non-conservative form. You can attempt to find the eigenvalues from this system, but do not spend long doing this, just admire the equation you need to simplify.

(c) By using primitive variables show that the 1D equations can be written

$$\frac{\partial}{\partial t}\begin{pmatrix}\rho\\ v_x\\ E\end{pmatrix}+\begin{pmatrix}v_x & \rho & 0\\ 0 & v_x & \alpha/\rho\\ 0 & (\alpha+1)E & (\alpha+1)v_x\end{pmatrix}\frac{\partial}{\partial x}\begin{pmatrix}\rho\\ v_x\\ E\end{pmatrix}=0$$

(d) For what values of α is the system hyperbolic?